



OPEN Optimization of connectome weights for a neural network model generating both forward and backward locomotion in *C. elegans*

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Previous studies tracking the relationship between manipulations of *C. elegans* neurons and the resulting behavioral changes have called for the development of a connectome-constrained neural network model that describes the cascade from neurons to behavior. However, the model using anatomical connectome weights directly did not achieve that. Here, we introduce a framework that jointly optimizes synaptic weights with respect to the relative proportions of anatomical synaptic weights, muscle activity patterns required for locomotion, and the phase relationship between SMD neuron activity and neck-bending angle. As a result, our neural network model generates plausible *C. elegans* behavior mediated by activity changes in forward and backward command-neurons, even without the introduction of pacemaker neurons with intrinsic oscillatory activity. Additionally, we identified necessary neurons for maintaining oscillatory patterns on muscular activity that could serve as clues for the central pattern generator in our neural network model. Finally, we provide 10 optimized synaptic weight sets of *C. elegans* that reproduce the results of manipulation experiments on the SMD neurons. This study will facilitate future studies for unraveling the multiscale relationship of “from synapse to behavior” in the nervous system.

Keywords *C. elegans*, Neural network simulation, Neural dynamics, Synapse-to-behavior, Machine-learning

The whole-animal anatomical connectome of *Caenorhabditis elegans* (*C. elegans*) provides an essential structural blueprint. Nevertheless, comparatively few modeling frameworks jointly leverage connectome-derived, quantitative synaptic weight information to bridge neuronal dynamics, muscle activity, and locomotor output within a mechanistic cascade. Reproducing a biological neural network of *C. elegans* therefore requires simulating experimentally observable characteristics that span both structure and function. Structural information of the neural network describes the connectome itself, entailing the measurements of the spatial dimensions of each synapse and contact volume between neurons. Functional information of the neural network captures the dynamical activity of neurons and other somatic cells, typically measured as time series under various experimental conditions, such as membrane potential recordings or calcium imaging. A model capable of reproducing both the structure and function of the neural network provides principles toward reconstructing a biologically grounded neural system. Additionally, it is crucial to elucidate how these structural and functional aspects of neural network are interconnected, based on the fundamental principles of neuroscience.

To tackle this challenge, *C. elegans* offers an ideal system given its compact nervous system of 302 neurons. Thus, neuroscience has long been driven to digitally replicate the neural network of *C. elegans*. As a hallmark animal model with fewer neurons than other animal models, such as fruit flies or mice, it presents heightened expectations for successful replication of the neural network. Furthermore, *C. elegans* is widely recognized for its importance not only in neuroscience but also throughout biological sciences. Therefore, there has been an ambition to reverse-engineer the neural network of *C. elegans* by creating a digital strategy that operates identically to the real *C. elegans*, aiming to produce a virtual animal for indefinite experimentation without the constraint of mortality. Realizing this ambition requires a connectome-constrained model in which the anatomical connectome provides the structural scaffold, while synaptic weights are inferred as effective coupling parameters to reproduce measurable neuronal and muscular dynamics that generate behavior.

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Previous studies have elucidated the connectome of *C. elegans*, revealing its whole-animal connectome structure, encompassing 302 neurons and 165 somatic cells. This milestone was achieved through the pioneering works of dedicated scientists over recent decades^{1–4}. Additionally, there have been further attempts to elucidate the connectome, including studies on variations across life stages⁵ and the Dauer state⁶, with additional experimental insights^{7,8}. As revealed by these studies, the connectome has become a crucial determinant for understanding the structure and functions of *C. elegans*’ neuronal circuits, including gap junctions and chemical synapses^{4,9}. Meanwhile, various studies have been conducted to decipher the operation of the neural network by measuring the calcium activity in numerous neurons of *C. elegans*^{10–17}. Recent research utilizing NeuroPAL technology¹⁸, which facilitates the identification of numerous neurons, has shed light on the functional excitatory-inhibitory relationships among neurons by simultaneously measuring their calcium activities in *C. elegans*¹⁹.

Several studies have aimed to simulate the neural network of *C. elegans* based on the anatomical connectome in which synaptic weights are the structural weights of anatomical contact volume between neurons, proposing various models to explain the neural network’s operation for locomotion^{20–25}. However, prior studies have not addressed this problem by optimizing synaptic weights through minimization of a combined objective function designed to jointly capture anatomical connectome structure, muscle activity patterns, and a specific experimentally observed neuronal activity feature, subject to biologically plausible constraints. Previous studies have either solely utilized the network topology of the anatomical connectome²⁵ or employed it with modifications for the repetitive unit structure of a neuronal circuit^{20,23,24}, or optimized connectome-constrained models primarily against a single posture-level objective function²⁷. Some research has criticized the direct usage of anatomical connectome weights for its limitations in explaining the neuronal activities of *C. elegans*^{11,17,19}. Other research^{21,22} has incorporated the anatomical connectome⁸ as-is, generating a central pattern in the simulation, but accounted for muscle cell activities by multiplying the projection coefficients of neuronal modes by eigenworm modes, principal components of body posture²⁶. This approach, while innovative, was insufficient for explaining neural-to-body translation within a unified dynamical formalism linking neuronal signaling to muscle activity. Thus, limitations of previous methods include relying only on connectome topology or modified circuit structure, using anatomical synaptic weights without inferring effective functional couplings, and employing neural-to-body eigenmode projection mappings that are not embedded in a unified dynamical formalism linking neuronal signaling to muscle activation.

Here, we propose that synaptic weight optimization of a combined error function addresses these limitations by relying on a fixed anatomical scaffold defined by connectome topology and connectivity, using anatomical synaptic weights as structural constraints for inferring effective functional couplings, and employing a shared leaky-integrator dynamical formalism for neuronal, muscular, and neuromuscular dynamics (Fig. 1). Thus, synaptic weight optimization of a combined error function serves as a principled mechanism for estimating effective functional couplings that reconcile a fixed anatomical scaffold with observable neuronal and muscular dynamics under multiple experimental constraints. In this framework, we jointly optimize the synaptic weights in the *C. elegans* artificial neural network (CANN) model and show that the resulting CANN models capture three targeted features within biologically plausible constraints, which are the relative proportions of anatomical synaptic weights, muscle activity patterns required for locomotion, and the phase relationship between SMD neuron activity and neck-bending angle. Furthermore, when the body control angles derived from the CANN model were input into the *C. elegans* body mechanical simulator²⁸, the resulting simulations yielded locomotion patterns qualitatively similar to published *C. elegans* kinematic patterns under the chosen output-to-angle

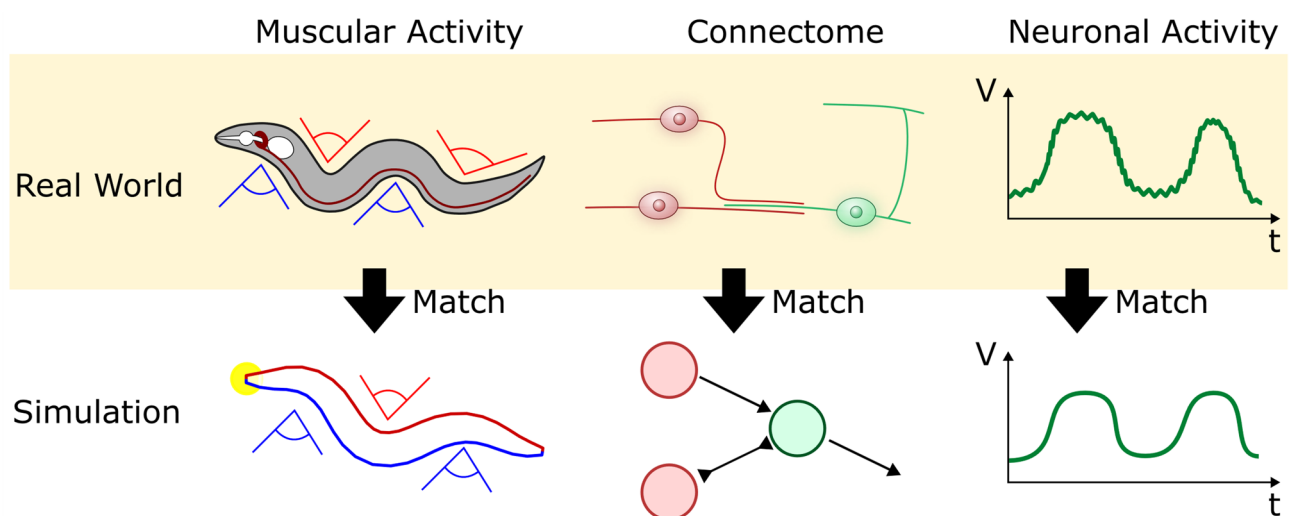


Fig. 1. Schematic diagram of the multiple objectives used for synaptic-weight optimization in a *C. elegans* neural network model. The schematic diagram illustrates the three objectives used for synaptic-weight optimization: the muscle-activity patterns required for locomotion (left), the relative proportions of anatomical synaptic weights (middle) and the phase relationship between SMD neuron activity and neck-bending angle (right).

mapping. Beyond the intended machine-learning objectives of the error function, some machine-learning trajectories of our model yielded locomotor phenotypes for transient receptor potential cation channel-1,2 (*trp-1,2*) mutants and SMD(D/V) ablation that were consistent with the experimental observations²⁹. Additionally, the necessary neurons for maintaining the oscillatory activation pattern of body-wall muscle cells were identified in our model. This study serves as a foundation for developing the connectome-constrained neural network model by linking the anatomical connectome, neuronal activities, and muscular activities that induce the behavior of *C. elegans*. It aims to reverse-engineer the biological neural network of *C. elegans* by considering its function, including neuronal activities and various behaviors, and may serve as a new reverse-engineering concept to elucidate the intrinsic working principles of the nervous system.

Materials and methods
Model overview and design principles

To clarify the design principles, Table 1 summarizes the model configurations, including constraints, tuned parameters, and data sources, distinguishing them from the validation metrics and the analysis used to evaluate emergent properties.

Acquisition of anatomical connectome data

We obtained the topology and synaptic weights of anatomical connectome from prior work². In the anatomical connectome data provided, we utilized information only on the connections between 95 body-wall muscle cells and 162 motor neurons linked to these muscles through synapses, as well as 10 pre-motor interneurons, which were referred to as command-neurons for locomotion (Supplementary Data 1). Thus, we considered a total of 2,291 chemical synaptic connections and 626 gap junction connections. All the anatomical weight of synapses are positive numbers, meaning none of them are anatomically absent. The polarity of synapses was set such that the chemical synapses of D-type motor neurons, which correspond to GABAergic neurons, were designated as inhibitory, while those of all other neurons were considered excitatory (Supplementary Data 2). Although recent neurotransmitter prediction studies suggest a genetic landscape of excitatory and inhibitory connections based on both neurotransmitter and receptor types, the majority of the synapses from D-type motor neurons in the ventral nerve cord is undetermined³⁰. Thus, we utilized neurotransmitter predictions as the primary basis for synaptic polarity assignment, adhering to the established view that D-type motor neurons in the ventral nerve cord function as inhibitory motor neurons^{4,31}. This study adopts the standard *C. elegans* neuron and muscle cell naming convention used in the anatomical data of previous work². Throughout the manuscript, SMD refers to one of the *C. elegans* head motor neuron classes, with SMDD and SMDV denoting the dorsal and ventral subclasses, respectively. BWM indicates body-wall muscle. In BWM labels, the d/v prefix indicates dorsal or ventral, the L/R suffix indicates left or right, and the terminal number denotes the position index along the body axis from head to tail.

Supplementation of proprioceptive connections into the connectome

Given the absence of proprioception information in the anatomical connectome data², we supplemented the CANN model's motor neurons with connections that facilitate the reception of proprioceptive signals, as detailed in previous studies^{29,32,33}. These supplemented proprioceptive connections are phenomenological feedback couplings introduced for modeling, not anatomically identified synapses. First, additional chemical synapses were introduced to the CANN model between the body-wall muscle cells and A/B-type motor neurons. This modification enables the neurons to receive proprioceptive signals from ventral body-wall muscle cells. The

Category	Description
Model Configuration	Topology and weights of anatomical connectome were acquired from previous work ²
	The membrane potential time constant was derived from prior measurements ³⁸
	Command-neuron membrane potentials were fixed to 0 or 1 depending on the locomotion state (forward/backward)
	Neuronal dynamics were governed by a leaky-integrator equation
	Parameters of the sigmoid function for synaptic signal transduction were held constant
	Synaptic polarity was set by neurotransmitter identity (GABA: inhibitory; others: excitatory)
	Proprioceptive connections absent from the anatomical connectome were added for A-/B-type motor neurons and SMD neurons based on prior work ^{11,29,32–35}
	The error function was designed to jointly enforce (1) muscle activity patterns, (2) the relative proportions of anatomical synaptic weights, (3) weight regularization, and (4) the phase relationship between SMD neuron activity and neck bending
	Coefficients for error terms were empirically tuned to control trade-offs among competing objectives during optimization
Parameter Fitting	The optimization converged to solutions that achieved the objectives: (i) strong positive correlation between optimized and anatomical synaptic weights, (ii) sinusoidal body-wall muscle activity, and (iii) SMD neuron activity pattern correlated with neck bending
	A mechanical body simulator ²⁸ was used to generate locomotion tracks and kinematic metrics (e.g., speed and wavenumber) from the CANN outputs
Out-of-objective Validation	Reproduced ventral bias in <i>trp-1,2</i> mutants via proprioception blocking and dorsoventral biases upon SMD ablation
Analysis	Type I/II necessary neurons for generating oscillatory activity required for locomotion were identified via in silico neuron elimination

Table 1. Summary of model configuration, parameter fitting, out-of-objective validation, and analysis. This table clarifies which elements were fixed, which were fit during optimization, and which findings were evaluated independently.

somas of A/B-type motor neurons, whose locations are detailed in previous work³⁴, are located approximately 0.12 mm apart from the somas of the muscle cells along the anterior–posterior axis. The displacements from the somas of neurons to those of muscle cells are further supported by the findings in previous studies^{33,35}. Second, new chemical synapses were introduced in the CANN model between the SMD motor neurons and the four body-wall muscle cells (dBWML4, dBWML4, vBWML4, and vBWML4; dorsal/ventral body-wall muscle, left/right, position 4), enabling the SMD motor neurons to receive proprioceptive signals related to the neck-bending angle^{11,29}. This modification is supported by previous research, which indicates that the neck-bending angle is in phase with the Ca^{2+} activity in SMDD neurons and antiphase with the Ca^{2+} activity in SMDV neurons²⁹. To further validate this fact, we employed a feature selection technique³⁶ on data capturing both dorsoventral body angles and SMD neuron activities simultaneously¹¹. This analysis revealed that the variance in SMD neuron Ca^{2+} activity is most strongly associated with the fourth dorsoventral body angle. Consequently, we hypothesized the presence of proprioceptive connections from the fourth body-wall muscle cells (dBWML4, dBWML4, vBWML4, vBWML4) to SMD neurons, connections that were not previously identified in anatomical studies. Here, we considered a total of 316 proprioceptive connections between neurons and muscles, with the polarity of these connections set as excitatory (Supplementary Data 2).

Time constant estimation of the membrane potential response

The time constant of the membrane potential response to an input current is one of the key components that determines the ability of a neuron to compute incoming signals³⁷. From the published experimental data³⁸, we estimated the time constant (τ) as 0.019 s for the RIM, AFD, AIY, and AWA (standard *C. elegans* neuron class names; see²) neurons by curve-fitting the membrane potential time series (Fig. S1). Detailed estimation procedure is provided in the Supplementary Information (§S1). The neuron time constants reported vary widely across studies: about 0.1 s³⁹, 0.002 s²⁰, 0.01 s²¹, and between 0.5 to 2 s²³. Our estimated time constant 0.019 falls within this range.

Leaky-integrator equation for membrane potential calculation

The leaky-integrator equation has been used to calculate the membrane potential of each somatic cell in *C. elegans*. This is because the leaky-integrator equation effectively explains the capacitance phenomenon of the cell's phospholipid bilayer and the ion leakage through leak channels, making it useful for calculating cell membrane potential⁴⁰. Previous studies have used the leaky-integrator equation to simulate the neural network of *C. elegans*. There are slight variations in the leaky-integrator equation among previous studies^{20,21,23,25,39–41}. We have also developed a version of the leaky-integrator similar to those found in prior research (See §S2). The leaky-integrator equation used in this paper is as follows:

$$\frac{dV_i}{dt} = \frac{1}{\tau} \left[(V_{\text{leak}} - V_i) + \sum_j w_{ij} r_j (V_{\text{chem},ij} - V_i) + \sum_j w'_{ij} (V_j - V_i) \right]$$

$$r_i = \frac{1}{1 + \exp(-\sigma(V_i - V_{\text{half}}))}$$

, where $\sigma = 5.9$, $V_{\text{half}} = 0.5$. t represents time. V represents the normalized membrane potential, a value between 0 and 1 in this study (See §S3). Specifically, V_i represents the membrane potential of cell i , $V_{\text{leak}} (=0)$ represents the electrical potential difference due to the membrane's leak channel, and $V_{\text{chem},ij}$ represents the electrical potential difference due to the ion channel of the chemical synapse from presynaptic neuron cell j to postsynaptic cell i . For an excitatory chemical synapse, $V_{\text{chem},ij}$ equals 1; for an inhibitory one, it equals 0. τ represents the time constant. The parameter w_{ij} denotes the synaptic weight of the chemical synapse from cell j to cell i ($w_{ij} \geq 0$), whereas w'_{ij} denotes the synaptic weight of the gap junction between cell i and cell j ($w'_{ij} \geq 0$). The parameters w_{ij} and w'_{ij} are constants that are independent of time t . The variable r_i represents the ion channel opening rate of the chemical synapse from cell i ; when cell i is a muscle cell, r_i also indicates the degree of muscle contraction. In reality, the degree of muscle contraction depends on the frequency of muscle action potentials^{42–44}, but this simplified abstraction is adopted in our model. There are lower and upper limits for both ion channel opening rate and degree of muscle contraction ($r_i \in (0, 1)$).

Configuration of the membrane potential of command-neurons

As in previous research²⁵, we ensured that the forward-locomotion command-neurons (AVB, PVC) and the backward-locomotion command-neurons (AVA, AVD, AVE) alternately had opposing membrane potentials^{12,17,31,45,46}. A forward-locomotion command is a state in which the membrane potentials of forward-locomotion command-neurons are 1, and those of backward-locomotion command-neurons are 0. Conversely, a backward-locomotion command is a state in which the membrane potentials of forward-locomotion command-neurons are 0, and those of backward-locomotion command-neurons are 1. When solving the leaky-integrator equation, the command at each time point t was determined to be either a forward- or backward-locomotion command.

Numerical method for solution of leaky-integrator equation

As the complexity of the neural network increases, the leaky-integrator equation becomes increasingly susceptible to stiffness issues of numerical integration⁴⁷. We employed the semi-implicit Euler method for numerical integration to resolve the stability issues of solving ordinary differential equations^{37,47}. To correct

inaccuracies in the decay rate of membrane potential, we employed a modified version of the semi-implicit Euler method with a time step (Δt) of 0.01 s (See §S4).

Definition of the error function

Inspired by research on methods for optimizing multiple objective functions jointly⁴⁸, we designed the combined form of the error function E to ensure that the CANN model could explain a variety of *C. elegans* experimental data^{2,29,49}.

$$E = E_{\text{BWM}} + \alpha_{\theta_w} E_{\theta_w} + \alpha_{\theta_{w'}} E_{\theta_{w'}} + \alpha_{|w|} E_{|w|} + \alpha_{\text{SMD}} E_{\text{SMD}}$$

E_{BWM} penalizes deviations of body-wall muscle activity from the sinusoidal locomotion pattern. E_{θ_w} and $E_{\theta_{w'}}$ encourage the optimized synaptic weights to align with the relative proportions of the anatomical connectome for chemical synapses and gap junctions, respectively. $E_{|w|}$ regularizes optimized synaptic weight magnitudes by discouraging growth toward unrealistically large synaptic strengths during optimization. E_{SMD} enforces the experimentally observed phase relationship between SMD activity and the neck-bending angle. Detailed formulation of each error term can be found in the Supplementary Information (§S5). α_* denotes the coefficient of each error term. By minimizing the error function E , we obtained the optimized synaptic weights w_{ij} and w'_{ij} , as well as initial membrane potentials $V_i^{(0)}$ for each neuron ($*^{(t)}$ denotes the value of a time-dependent variable at time t). This multi-objective optimization constrains the CANN model to remain consistent with the anatomical connectome while generating muscular and neuronal activity required for locomotion, consistent with the experimental observations. Throughout the optimization, the predefined connectivity topology was kept fixed and only the magnitudes of existing synaptic weights were updated with the minimum value of 0.001.

Error function optimization

Because the objectives compete, the coefficients α_* were empirically tuned to control the trade-off among error terms and to avoid simulation collapse or severe degradation of individual objectives during optimization. The coefficients were set to $\alpha_{\theta_w} = 0.5$, $\alpha_{\theta_{w'}} = 4$, $\alpha_{|w|} = 0.1$, and $\alpha_{\text{SMD}} = 1$. For the time-series-dependent terms, α_{SMD} was chosen based on per-cell error contributions of E_{BWM} and E_{SMD} , whereas for E_{θ_w} , $E_{\theta_{w'}}$, and $E_{|w|}$, the corresponding coefficients were chosen considering that these terms act directly on synaptic weights and thereby influence the Jacobian of the leaky-integrator dynamics and thus the full temporal evolution of the leaky-integrator dynamics. Using these coefficients α_* , the minimization process of the error function E is as follows. First, w_{ij} and w'_{ij} were initialized with random values, where w_{ij} ranges from 0.001 to 2, and w'_{ij} from 0.001 to 1. The membrane potential at $t = 0$ was initialized as $V_i^{(0)}$. The leaky-integrator equation was numerically solved to obtain V_i and r_i as a function of time t (See §S6). The value of the error function E was computed from V_i and r_i over time t , together with w_{ij} and w'_{ij} . The Adam optimizer⁵⁰ was employed to update w_{ij} , w'_{ij} , and $V_i^{(0)}$ to minimize the error function E , with the learning rate set to 0.001, and the Adam parameters set to $\beta_1 = 0.9$ and $\beta_2 = 0.999$ (See §S7). We created and utilized an adaptive error minimization method to reduce the computational cost of the error minimization process (See §S8).

Stepwise neuron elimination and single neuron elimination

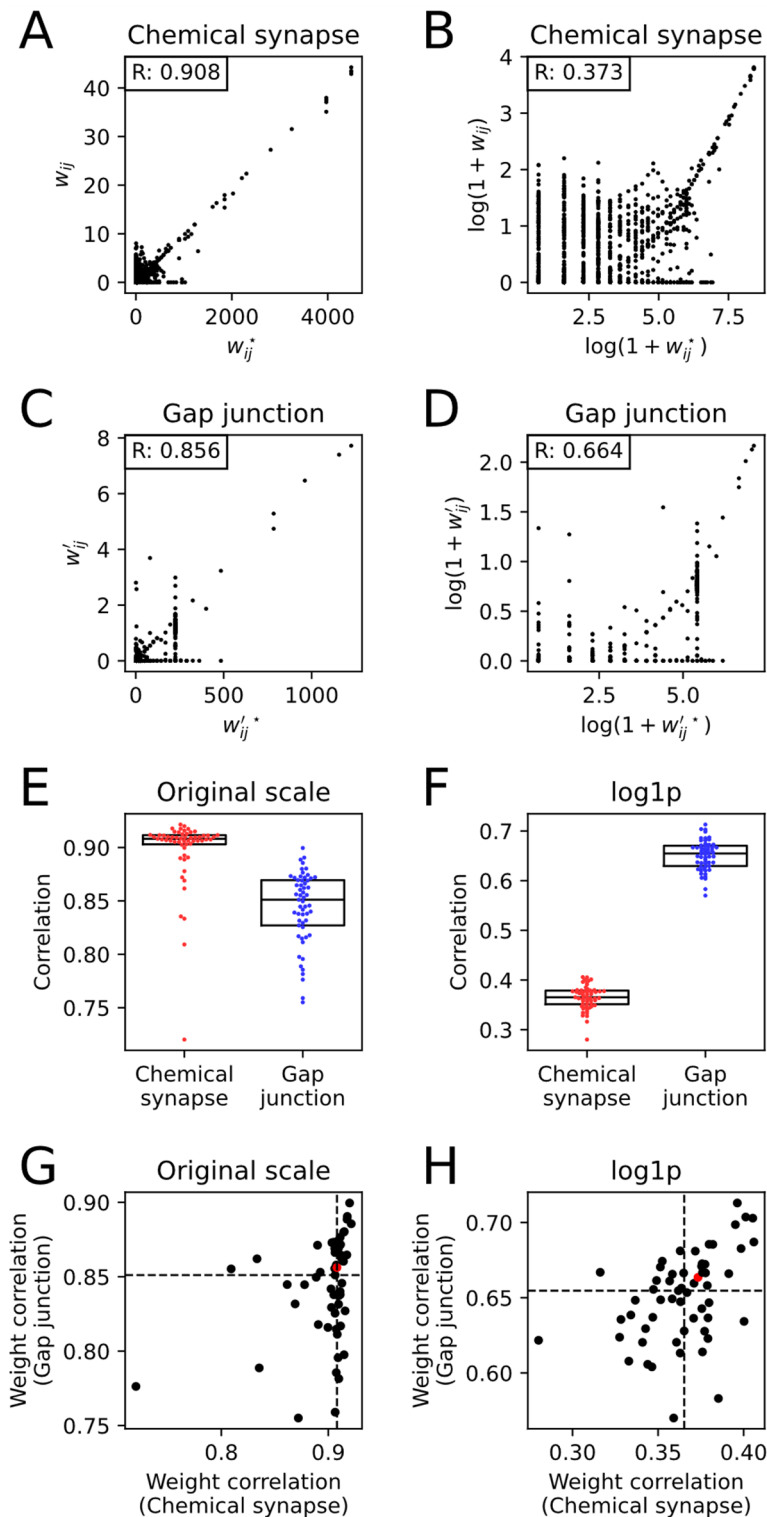
Stepwise neuron elimination identifies a set of type I necessary neurons required to maintain the oscillatory activation pattern of body-wall muscle cells, which is essential for forward and backward locomotion. Neurons in the CANN were iteratively ablated one at a time, starting with the neurons that had the least impact on the oscillatory muscle activation pattern (Fig. S2A). Neurons in left-right symmetry were treated independently within the iterative ablation process. Starting from the intact CANN where no neurons were ablated, we computed the membrane potential time series ($V_{i \in \text{BWM}}^{(t)}$) of the 95 body-wall muscle cells for each case of single additional neuron ablation (Fig. S2B). To assess whether a given ablation abolished muscle activity oscillation, we applied dynamic mode decomposition (DMD)⁵¹ to the muscle membrane potential time series. DMD decomposes a high-dimensional time series into a set of spatio-temporal modes, each characterized by an oscillation period (T) and a decay factor (γ), where larger γ values indicating slower decay of the mode. We define A_4 as the magnitude of the temporal coefficient of the mode with the largest γ , evaluated at $t = 4\text{sec}$. We chose this time point to reduce sensitivity to early transients. At each iteration, we selected the single additional ablation scenario that maximized A_4 , updated the network accordingly, and repeated the procedure until we obtained the smallest set of motor neurons that satisfied $A_4 > 0.1$ and $T < 4.5\text{sec}$. The period criterion ensures at least two oscillation cycles within 10-s time window of muscle activity time series. We defined this minimal set of motor neurons as the type I necessary neuron group.

We defined a type II necessary neuron as a motor neuron that caused the disappearance of the muscle membrane potential oscillations when ablated in isolation. Oscillations were considered absent when $\gamma < 0.1$, $T \leq 0\text{ sec}$, or $T \geq 4.5\text{ sec}$.

Notably, the command interneurons were excluded from the type I/II necessary neuron identification since they were not treated as dynamical variables but were instead assigned externally imposed binary command states.

Analysis code and tools

The analyses were conducted using custom Python code⁵² based on NumPy⁵³, and Numba⁵⁴. Visualizations were created with Matplotlib⁵⁵. The mechanical body simulator of *C. elegans* (ElegansBot) was used for the locomotion simulation²⁸. ElegansBot is a two-dimensional rigid body chain model that simulates *C. elegans* locomotion by applying Newtonian equations of motion to each body segment. ElegansBot reproduces various locomotive behaviors and quantifies force distributions, enabling the study of muscle-neuron interactions and environmental



friction effects. To interface the CANN output with ElegansBot, the muscle contraction variable ($r_i^{(t)}$, Fig. S3) was converted to body control angles ($\theta_{ctrl,k}^{(t)}$):

$$\theta_{ctrl,k}^{(t)} = 1.6 \times \frac{1}{4} \frac{r_{[dBWMLk]}^{(t)} + r_{[dBWMLk]}^{(t)} - \tilde{r}_{[vBWMLk]}^{(t)} - r_{[vBWMLk]}^{(t)}}{r_{on} - r_{off}}$$

where the superscript (t) denotes a time point t and the factor 1.6 is a scaling factor whose value is arbitrarily determined. $\tilde{r}_{[vBWMLk]}^{(t)}$ is a linearly interpolated value of $r_{[vBWMLk]}^{(t)}$, adjusted to transition from 23 evenly spaced points to 24 evenly spaced points throughout the length of the *C. elegans*. This conversion includes a

Fig. 2. Synaptic weights comparison between anatomical connectome and CANN model. (A, B, C, D) Scatter plots comparing anatomical synaptic weights to optimized synaptic weights of a representative machine-learning trajectory. Only synapses present in the anatomical connectome are included in the comparisons. The number in the top left corner of each box indicates the correlation coefficient (R). (A, B) Chemical synapse in original scale and log1p transformed scale. Here, log1p indicates $\log(1 + x)$ function, where log denotes the natural logarithm. (C, D) Gap junction in original scale and log1p-transformed scale. (E, F) Correlation value between anatomical synaptic weights and optimized synaptic weights for each machine-learning trajectory for original weight scale and log1p-transformed weight scale. Each point represents a machine-learning trajectory ($n = 57$ per condition; red/blue). The y-axis component of each point indicates the correlation value (red for chemical synapse, blue for gap junction). The x-axis component of each point does not represent any information; it is determined solely to minimize overlap between points. Box plots indicate median and interquartile range (Q1–Q3). (G, H) Scatter plot of chemical synapse weight correlation and gap junction weight correlation for each machine-learning trajectory for original weight scale and log1p-transformed weight scale. Dotted lines represent the median value for each axis, and the red point denotes the representative machine-learning trajectory of (A, B, C, D).

constant scaling factor that was introduced to match the dimensionless muscle-activity output of the CANN model to the angle units required by ElegansBot. This scaling factor is not a biophysically derived parameter, but an interface parameter for kinematic readout and neuromechanical visualization.

Results

Optimized synaptic weights were positively correlated with anatomical weights.

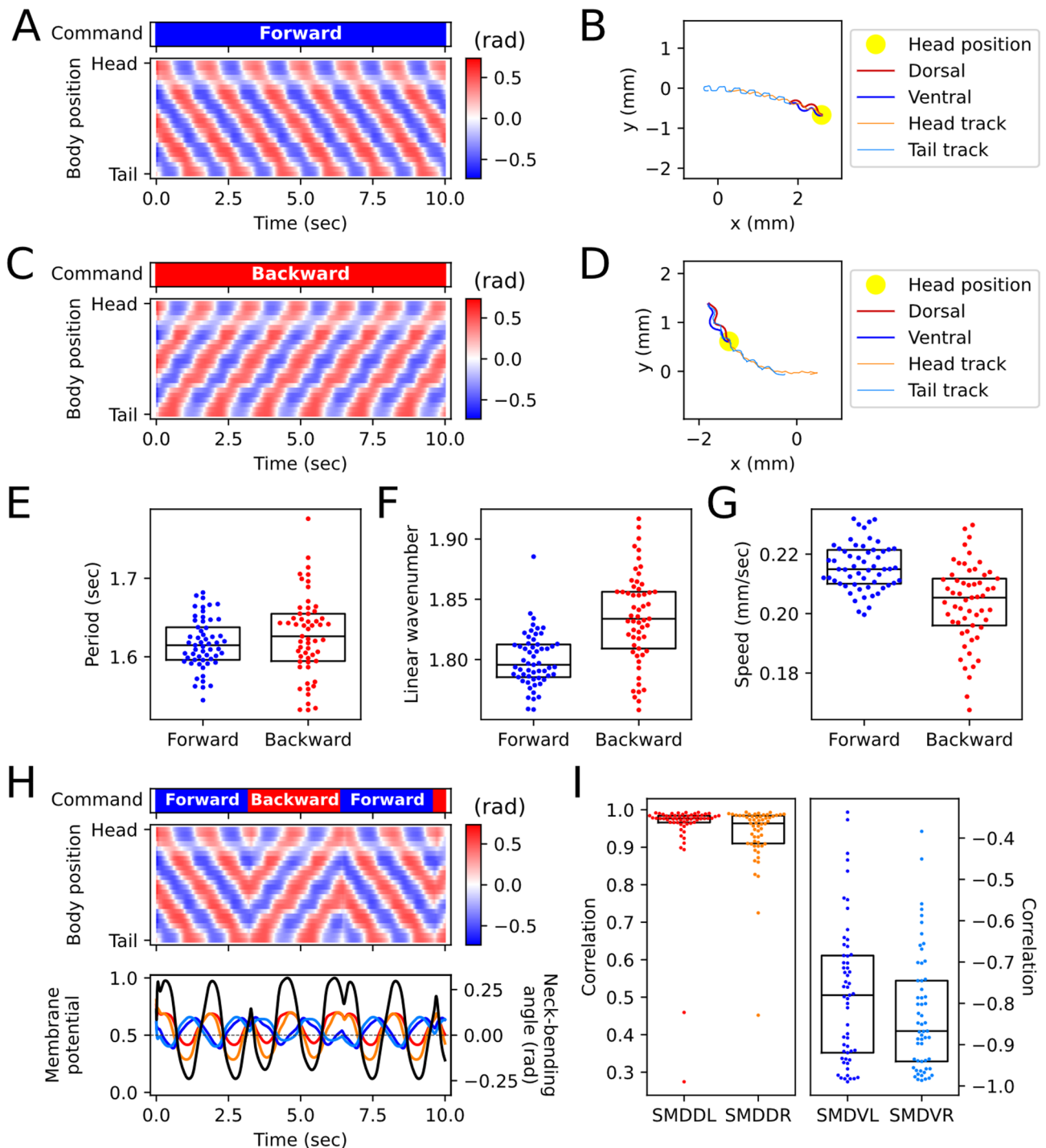
We developed a CANN model that controls *C. elegans* muscle cells for locomotion using a leaky-integrator equation. For the CANN model to generate appropriate synaptic currents, 3,233 synaptic weight values (w_{ij} and w'_{ij}) of connectomes must be accurately adjusted. We have developed a neural network model that generates the muscle membrane potentials necessary for locomotion by formulating an error function and minimizing the error function by adjusting synaptic weights (See Materials and Methods). Based on this random initialization setup, we ran 300 machine-learning trajectories, of which 57 resulted in a CANN model that produced the muscle membrane potentials necessary for forward and backward locomotion. Importantly, including multiple experimental data targets in the error function does not guarantee that all of them can be satisfied jointly: the constraints compete and define a narrow feasible region in parameter space. Consistent with this, only 57 out of 300 optimization trajectories converged (Fig. S4), indicating that achieving the baseline targets is itself non-trivial rather than automatic. Whereas $E_{|w|}$ primarily served to restrain the growth of synaptic-weight magnitudes, the other error terms were reduced during optimization, supporting that the empirically chosen coefficients enabled joint optimization without the process being effectively dominated by only one or two terms (Fig. S5).

For each of these 57 converged machine-learning trajectories, we calculated Pearson correlations between optimized and anatomical synaptic weights on both the original scale and the log1p-transformed scale (Fig. 2A–H). In this analysis, the supplemented proprioceptive connections were excluded, which were not present in the anatomical connectome. On the original scale, the median correlation among the machine-learning trajectories was 0.91 (Interquartile range, IQR: 0.01) for chemical synapses and 0.85 (IQR: 0.04) for gap junctions (Fig. 2E). After log1p transformation, the corresponding median correlations were 0.37 (IQR: 0.03) for chemical synapses and 0.65 (IQR: 0.04) for gap junctions (Fig. 2F). Thus, weight correlations in the original scale were tighter than those in the log1p-transformed scale, especially for chemical synapses, indicating that correspondence is strongest for larger-weight connections. There was no clear relationship between chemical synapse and gap junction correlation values across trajectories in either the original scale or the log1p-transformed scale (Fig. 2G, H).

Some optimized weights converged to values near zero, whereas some anatomically weak synapses were assigned relatively large optimized weights in the fitted locomotion regime of our model (Fig. 2A, C). These quantities should be interpreted as effective couplings within the simulation model. Notably, all anatomical synaptic weights are positive, indicating that all synapses included in the analysis were present in the anatomical dataset, even when their anatomical weights were very small. In addition, the magnitudes of optimized and anatomical synaptic weights differed systematically because the two quantities have different units. The anatomical connectome provides a structural measure of synapse size, whereas the optimized parameters represent effective couplings in the dynamical model. Having obtained these optimized synaptic weight sets, we next asked whether these weights also generate realistic kinematics when coupled to a biomechanical body model.

Forward and backward locomotor patterns and SMD phase relationships were generated by the CANN model with optimized synaptic weights.

Degrees of muscle contraction ($r_i, i \in \{\text{BWM}\}$) were calculated from a CANN model of each machine-learning trajectory under forward-locomotion command for 10 s of simulation time. The body control angles ($\theta_{\text{ctrl},k}, k \in \{1, 2, \dots, 24\}$) were derived from these degrees of muscle contraction (See Materials and Methods) and exhibited a sine wave pattern, resembling the kymogram pattern, a spatio-temporal body posture map, observed in real *C. elegans* (Fig. 3A). By inputting the body control angles into the *C. elegans* mechanical body simulator which calculates the thrust from muscle forces and environmental friction, we generated spatio-temporal track of forward locomotion (Fig. 3B; Supplementary Video 1; See Materials and Methods). Similarly, degrees of muscle contraction were calculated from a CANN model under backward-locomotion command, and the body



control angles were calculated from these degrees of muscle contraction and exhibited a sinusoidal kymogram during backward locomotion (Fig. 3C). Inputting the body control angles into the mechanical body model, the simulation showed backward locomotion track (Fig. 3D, Supplementary Video 2).

From body control angles of forward and backward locomotion, period and linear wavenumber of locomotion were derived for each machine-learning trajectory, by DMD analysis on the body control angles and sinusoidal curve fitting to the oscillating DMD mode pair of the longest period. (Fig. 3E, F). From the spatio-temporal track of locomotion, the mechanical speed was calculated for each machine-learning trajectory (Fig. 3G). The median period, the median linear wavenumber, and the median speed across machine-learning trajectories were 1.61 s (IQR: 0.04 s), 1.80 (IQR: 0.03), and 0.21 mm/sec (IQR: 0.01 mm/sec) for forward locomotion and 1.63 s (IQR: 0.06 s), 1.83 (IQR: 0.05), and 0.20 mm/sec (IQR: 0.02 mm/sec) for backward locomotion. When a real *C. elegans* crawls forward on an agar plate, the period, wavenumber, and speed are 1.6 s, 1.83, and 0.21 mm/sec, respectively²⁸. This suggests that the CANN model can generate control signals necessary for both forward and backward locomotion. Under a command alternating between forward- and backward-locomotion command every 3.2 s, with a total duration of 10 s, degrees of muscle contraction were generated from CANN model. The

◀ **Fig. 3.** Neuronal and muscular activity reproduced using CANN model. **(A)** Body control angles determined by the muscle membrane potential over time from CANN model of the representative machine-learning trajectory under forward-locomotion command. Upper: Locomotion command (blue: forward-locomotion command, red: backward-locomotion command). Lower: Body control angles. **(B)** The locomotion track derived from the input body control angles of **(A)**. The yellow circle indicates the head position of *C. elegans*. The orange and sky-blue lines mean head and tail tracks, respectively. **(C)** Body control angles over time determined by the muscle membrane potentials from CANN model of the representative machine-learning trajectory under backward-locomotion command. **(D)** The locomotion track derived from the input body control angles of **(C)**. **(E)** Oscillation period calculated from body control angles. Each point represents a machine-learning trajectory ($n = 57$ per condition; blue/red). The y-axis component of each point indicates the correlation value (blue for forward locomotion, red for backward locomotion). The x-axis component of each point does not convey any meaning; it is determined solely to minimize overlap between points. Box plots indicate median and interquartile range (Q1–Q3). **(F)** Oscillation linear wavenumber calculated from body control angles. **(G)** The locomotion speed resulted from input body control angles of each machine-learning trajectory. **(H)** Locomotion command (upper panel) triggered the CANN model to generate body control angles (middle panel), the membrane potential of SMD neurons (lower panel: red for SMDDL, orange for SMDDR, blue for SMDVL, sky-blue for SMDVR), and neck-bending angle (black), resulted from the representative machine-learning trajectory. **(I)** Pearson correlation between neck-bending angle and membrane potential of SMD neurons. Each point represents a machine-learning trajectory ($n = 57$ per condition; red/orange/blue/sky-blue).

body control angles, calculated from the degrees of muscle contraction, showed temporally alternating forward and backward locomotion patterns. To check whether membrane potential of SMD neurons calculated from CANN model and neck-bending angle has in-phase or anti-phase relationship, we compared the membrane potential of SMD neurons and the neck-bending angle ($\theta_{\text{ctrl},4}$) for each machine-learning trajectory (Fig. 3H, Supplementary Video 3) by calculating Pearson correlation between them (Fig. 3I). The median correlation among machine-learning trajectories was 0.98 (IQR: 0.02) for SMDDL, 0.96 (IQR: 0.07) for SMDDR, -0.8 (IQR: 0.2) for SMDVL, -0.9 (IQR: 0.2) for SMDVR. These results were consistent with the biological experiments that showed that the Ca^{2+} activity of the SMDD neuron was in phase with the neck-bending angle, and the Ca^{2+} activity of the SMDV neuron was in antiphase with the neck-bending angle^{11,29}.

In summary, the CANN model with the optimized connectomes derived in this study generated forward and backward locomotion-related activity patterns and SMD neuronal membrane-potential patterns that were consistent with the targeted experimental observations. Thus, we developed a CANN model that captured the main targeted features of our framework (Fig. 1). When coupled to the mechanical body model, the resulting control signals generated spatio-temporal locomotion patterns in *ElegansBot* under the chosen output-to-angle mapping.

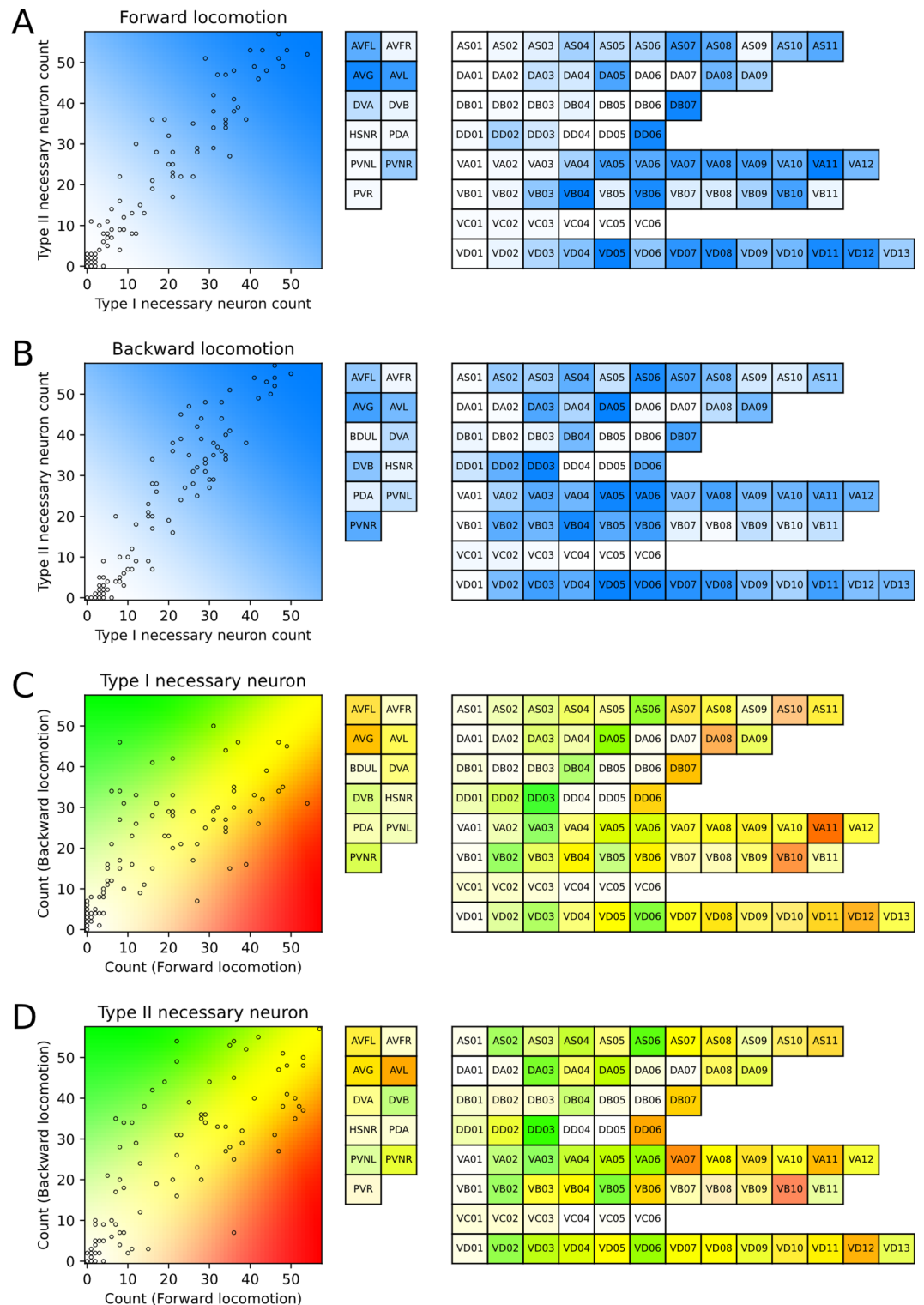
The type I/II necessary neurons that sustain the periodic activation of *C. elegans* body-wall muscle cells have been identified.

Having established that the CANN model successfully generates locomotion, we next sought to identify the minimal neuronal substrates underpinning these rhythmic patterns. *C. elegans* produces sinusoidal wave patterns during locomotion. For such sinusoidal wave patterns to exist, a central pattern generator (CPG) that produces oscillatory signals must be present within *C. elegans*. Previous studies^{32,56–58} have provided hypotheses and experimental evidence that the CPG of *C. elegans* resides within the neuronal circuit and is facilitated by proprioception. In this study, we present results identifying the neuron groups that are necessary for sustaining the oscillatory activation of muscle cells, which may provide clues about the *C. elegans* CPG. This was achieved using ablation-based methods, including stepwise neuron elimination or single neuron elimination (Methods).

The type I necessary neuron group identified through stepwise neuron elimination represents the minimal set of motor neurons predicted to be necessary for generating muscle membrane potential oscillation. The type II necessary neuron is a neuron whose removal from an intact CANN model eliminates muscle membrane potential oscillation. Among the 57 machine-learning trajectories that have learned both forward and backward locomotion, differences were observed in the neuron configuration of the type I necessary neuron group. Consequently, we counted the number of times each neuron appeared as a member of the type I necessary neuron group across the 57 machine-learning trajectories. Similarly, we counted how often each neuron appeared as a type II necessary neuron across the 57 machine-learning trajectories. Neurons that appear more frequently can be considered more crucial for generating oscillatory membrane potentials in muscles within the CANN model.

Type I necessary neurons for forward locomotion and those identified as type II showed comparable counts across trajectories (Fig. 4A). To prevent shared zeros from inflating the Pearson correlation, we excluded low-count neurons, defined as neurons observed in fewer than 10 of the 57 trajectories in both counts for a given correlation pair. After this filtering, the Pearson correlation between the counts of type I and type II necessary neurons for forward locomotion was 0.88 (all neurons: 0.94). A strong correlation between the two counts indicates that neurons with high counts are both type I and type II necessary neurons for maintaining the oscillatory membrane potential pattern in muscles within the CANN model. Similarly to the forward locomotion scenario, when excluding low-count neurons, the Pearson correlation between the counts of type I and type II necessary neurons for backward locomotion was found to be 0.83 (all neurons: 0.96) (Fig. 4B).

We also compared the type I necessary neuron counts between forward and backward locomotion (Fig. 4C). When excluding low-count neurons, the correlation between the type I necessary neuron counts for forward



and backward locomotion was 0.46 (all neurons: 0.82). A relatively weak correlation between the two counts suggests that the composition of neurons maintaining the oscillatory membrane potential pattern in muscles within the CANN model may differ between forward and backward locomotion. The average number of type I necessary neurons was 27.7 for forward locomotion and 34.8 for backward locomotion, respectively. A similar relationship was observed in the type II necessary neuron counts. When excluding low-count neurons, the correlation between the type II necessary neuron counts for forward and backward locomotion was 0.52 (all neurons: 0.78) (Fig. 4D). The average number of type II necessary neurons was 33.7 for forward locomotion and 38.7 for backward locomotion, respectively. Within this model, backward locomotion was associated with higher type I and type II necessary neuron counts than forward locomotion.

◀ **Fig. 4.** Identification of type I/II necessary neuron sets for maintaining oscillatory patterns on muscular activity. **(A)** Type I necessary neuron counts versus type II necessary neuron counts for forward locomotion. Each hollow dot in the left diagram represents a motor neuron. The x-coordinate of the hollow dot indicates the number of machine-learning trajectories for which the motor neuron is selected as a type I necessary neuron, and the y-coordinate indicates the number of trajectories for which the motor neuron is selected as a type II necessary neuron for forward locomotion. The text in each cell of the right table indicates the neuron's name, and the color corresponds to the background color at the location of the corresponding hollow dot in the left diagram. If different motor neurons have the same type I necessary neuron counts and the same type II necessary neuron counts, the corresponding hollow dots for the motor neurons in the left diagram overlap. **(B)** type I necessary neuron counts versus type II necessary neuron counts for backward locomotion. This follows the same representation and methodology as described for forward locomotion in **(A)**. **(C)** Forward locomotion versus backward locomotion type I necessary neuron counts. This follows the same representation and methodology as described in **(A)**. A motor neuron in the table is more important to the forward locomotion if the color is redder, more important to the backward locomotion if the color is greener, and equally important to both forward and backward locomotion if the color is yellow. **(D)** Forward locomotion versus backward locomotion type II necessary neuron counts. This follows the same representation and methodology as described in **(C)**.

The impact of *trp-1,2* mutation and SMD(D/V) ablation observed in the proprioception experiments of *C. elegans* was reproduced.

To assess whether our CANN model reproduces the results of proprioception experiments from prior research²⁹, we calculated the body control angle from the muscle cell contraction generated by the CANN model under forward-locomotion command for a duration of 10 s and computed the locomotion track of *C. elegans* by inputting the body control angle into the mechanical body simulator²⁸ (Fig. 5A). For each of the 57 machine-learning trajectories which can perform both forward and backward locomotion, we calculated the degrees of muscle contraction from CANN model under forward-locomotion command and then calculated the body control angles based on these degrees of muscle contraction. We calculated the average body control angle over all time and angle index pair (t, i). If the absolute value of this average was above 0.01, it might not reveal the changes in locomotion caused by mutation or ablation, as wild-type *C. elegans* in these machine-learning trajectories performs circular locomotion. Therefore, we excluded 11 machine-learning trajectories with the absolute value of average body control angle above 0.01 and considered the remaining 46 machine-learning trajectories as candidate trajectories for reproducing the results of the biological experiments.

For each of the 46 candidate trajectories, we compared how locomotion differed between the unmanipulated case (wild-type) and the manipulated case (*trp-1,2*, SMD(D/V) ablation). We calculated the difference in body control angle by subtracting the body control angles of the unmanipulated case from that of the manipulated case. We calculated the average difference in body control angle over time and angle index pair (t, i), naming this as the mean angle difference (Fig. 5B). We defined the manipulated case as performing ventrally biased locomotion if the mean angle difference was below -0.01, and as performing dorsally biased locomotion if the mean angle difference was above 0.01.

The TRP-1,2 knockout mutant (*trp-1,2*) is known to exhibit ventrally biased locomotion. This is because the channel proteins expressed by the TRP-1,2 genes are involved in the neck-bending angle proprioception of SMDD neurons, and this proprioception is impaired by the *trp-1,2* mutation²⁹. To mimic the effects of this mutation, the proprioceptive connection between SMDD neurons and muscles was removed in the CANN model, blocking the proprioception signal. As a result, 15 out of the 46 candidate trajectories exhibited the ventrally biased locomotion (Fig. 5B).

When SMDD (or SMDV) neurons in actual *C. elegans* are ablated, it was known to result in ventrally-biased (or dorsally-biased) locomotion²⁹. To mimic the ablation of specific neurons in actual *C. elegans* within the CANN model, all synapses connected to the respective neurons were removed. We termed this action of removing all synapses connected to specific neurons in the CANN model as 'neuronal ablation' in the simulation. In biological experiments, 'ablation' refers to the destruction of specific neurons using lasers or cautery. For the total of 46 candidate trajectories, 17 trajectories exhibited dorsally biased locomotion when SMDD was ablated, and 26 trajectories exhibited ventrally biased locomotion when SMDV was ablated (Fig. 5B). Notably, 10 trajectories demonstrated both effects, showing the dorsally biased locomotion when SMDD was ablated and the ventrally biased locomotion when SMDV was ablated (Fig. 5C). These 10 trajectories also reproduced the effect of the *trp-1,2* mutation (Supplementary Video 4). In other words, since these 10 trajectories out of a total of 46 successfully reproduced the effects of the *trp-1,2* mutation and SMD(D/V) ablation, we evaluated the optimized synaptic weight sets of these 10 trajectories as the most promising candidate synaptic weight sets within our model class (Supplementary Data 3). When each of these 10 most promising candidate weight sets is applied to the CANN model, the membrane potentials of all cells included in the model can be observed during alternating forward and backward locomotion (Fig. S6).

Discussion

Multi-objective connectome-constrained optimization

Prior studies did not provide a unified framework for inferring synaptic weights while jointly fitting multiple locomotion-relevant experimental constraints in *C. elegans*. Nevertheless, several studies successfully reproduced specific experimental observations, each with distinct strengths and limitations^{22,23,25}. One model managed to perfectly replicate the topology and partially reproduce the kymogram, but the reproduction of anatomical

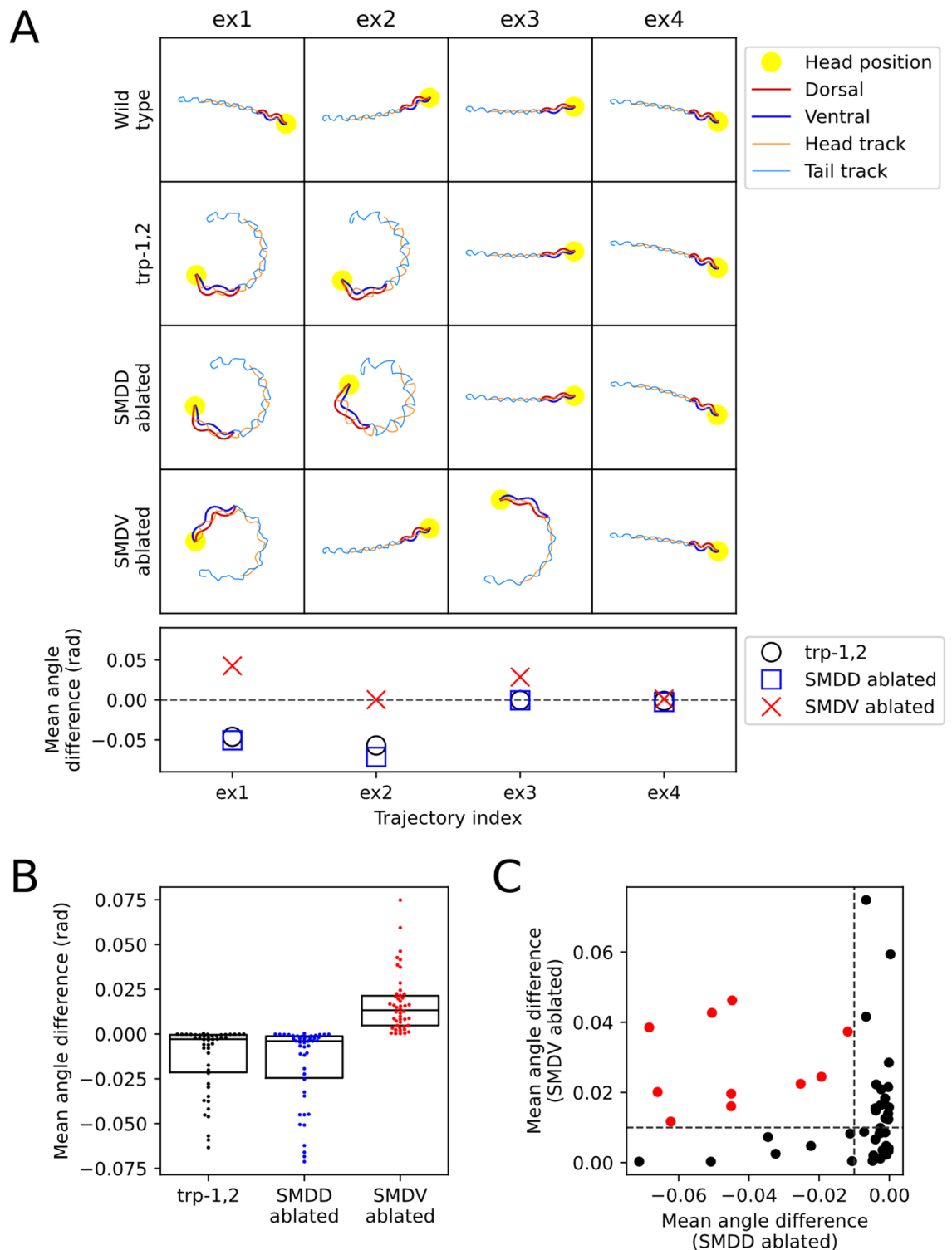


Fig. 5. Implementation of proprioceptive sensory restriction and neuronal ablation experiments using CANN model. (A) The impact of *trp-1,2* mutation (elimination of proprioceptive sensations of SMDD neuron) and SMD(D/V) neuron ablation on the locomotion track of the worm for each of four example machine-learning trajectories (ex1, ex2, ex3, ex4, each representing the index of a trajectory). Top: The locomotion track of each example machine-learning trajectory. Bottom: Mean angle difference. (B) Distribution of mean angle differences of candidate trajectories. Each point represents a candidate trajectory ($n = 46$ per condition; black/blue/red). Box plots indicate median and interquartile range (Q1–Q3). (C) Scatter plot of the mean angle difference for each candidate trajectory. Red dots represent candidate trajectories that match the locomotion phenotype caused by *trp-1,2* mutation and SMD(D/V) ablation, as reported in the biological experiment paper²⁹.

synaptic weights of the connectome was limited as the model only generated the distribution shape of synaptic weights similar to the anatomical connectome²⁵. Other models successfully reproduced the forward and backward crawling kymogram depending on the inputs from command-neurons, though the synaptic weights of the connectome were identified through computational search methods, leaving no guarantee or indication that the modeled synaptic weights would match anatomical synaptic weights^{23,24}. Moreover, the model had both excitatory and inhibitory synapses from a single SMD (or RMD) motor neuron. Yet another model used the anatomical connectome as-is and successfully reproduced central pattern generation²². However, the equation determining muscle activity to exhibit a sinusoidal pattern, by multiplying projection coefficients of neuronal modes to eigenmodes²⁶, was qualitatively different from the equation governing neuronal activity in the model. Thus, strategies focused on perfectly reproducing a single type of biological experiments may prove to be impractical. Instead, we adapted the error function of the previous model²⁵ to train it to simulate not only muscle activity patterns but also the connectome's synaptic weights and neuronal activity patterns. Consequently, our strategy aims not to optimize synaptic weights according to any single criterion, but to optimize them jointly with respect to the relative proportions of anatomical synaptic weights, the muscle activity patterns required for locomotion, and the phase relationship between SMD neuron activity and neck-bending angle.

By contrast, assigning a large coefficient to a single error term in our model's error function can lead the model to explain one type of experiment well while failing to account for others, similar to previous studies^{22–25}. Although the prior work²⁷ employed connectome-constrained optimization, their reliance on an exclusively posture-based error function limited the model's intrinsic oscillatory capabilities, forcing the model to rely on constant or pulsed external input to sensory neuron or forced input oscillation to make the model generate oscillatory activity output. This approach resulted in escape responses that do not match the tens-of-millisecond scale onset of touch-evoked reversals observed in vivo⁵⁹, as the simulation required a stimulus duration of 2 s. Additionally, the simulation presents physiological inconsistency in proprioception where RIV stimulation induces turning while SMDV ablation completely abolishes it, despite both neurons having neuromuscular junctions to ventral neck muscles. These are due to the difficulty in reducing each and every error terms to the desired level, as increasing the coefficient of one error term to reduce its value can result in a trade-off where other error terms increase. Nonetheless, our results support the practical utility of this multi-objective methodology, which jointly optimizes synaptic weights with respect to the relative proportions of anatomical synaptic weights, muscle activity patterns required for locomotion, and the phase relationship between SMD neuron activity and neck-bending angle.

Interpretation of the relationship between anatomical and optimized synaptic weights

While previous studies have noted that anatomical synapse counts do not always serve as a direct proxy for functional synaptic strength^{11,17–19}, our results suggest that, in a non-linear connectome-constrained neural network model, optimized weights remain associated with anatomical counts, supporting the use of anatomical connectivity to guide functional model fitting through a weighted objective. At the same time, the optimized weights in our model should be understood as model-dependent effective couplings rather than as direct estimates of biological synaptic strength. Accordingly, differences between anatomical and optimized synaptic weights do not imply that the anatomical measurements are uninformative. Instead, they likely reflect the gap between structural connectivity and the effective couplings required to reproduce locomotor function under the assumptions of our framework.

From this perspective, the optimized weights were not obtained by a simple global rescaling of the anatomical connectome. Instead, some anatomically identified connections were assigned smaller effective weights, whereas others were assigned larger effective weights. These deviations may reflect the model absorbing unrepresented physiological and dynamical factors into task-dependent effective couplings. In this sense, differences between anatomical and optimized synaptic weights may help identify where anatomical connectivity alone is insufficient to account for locomotor function within the present model and may provide candidate relationships for future experimental testing.

Emergent bidirectional locomotion without pacemakers

The CANN model, optimized using this multi-objective strategy, generated both forward and backward locomotion in *C. elegans* without the need for a preferred pacemaker neuron. In neural network models incorporating pacemaker neurons, these neurons autonomously generate oscillatory activation patterns, and the network outputs appropriate muscle contraction/relaxation patterns to produce forward and backward locomotion based on these intrinsic oscillations. However, there is still little evidence supporting the existence of pacemaker neurons specifically responsible for both forward and backward locomotion in *C. elegans*. Instead, it is well established that the on/off activity of certain command-neurons, such as AVA and AVB, triggers forward and backward locomotion. The CANN model captures the function of these command-neurons and demonstrates that reversing their activity can switch between forward and backward movement. The oscillatory membrane potential patterns generated in the CANN model rely heavily on proprioceptive connections that are organized and operated across A/B-type motor neurons and SMD neurons. The feedback loop between these neurons and muscle cells enables the generation of such oscillatory patterns. Furthermore, in the 10 most promising candidate CANN models that successfully replicated the effects of SMD(D/V) neuron ablation experiments, we observed that the removal of proprioceptive connection in SMD neurons (simulating the *trp-1,2* mutation experiment) produced circular locomotion in the model. This suggests that proprioceptive connections are important for maintaining the normal oscillatory pattern of muscle membrane potentials. The mechanism by which this perturbation leads to circular locomotion warrants further investigation, and a more detailed dynamical account will require future analysis. A related emergent feature of the model is the larger type I and type II necessary neuron counts observed during backward locomotion. One possible interpretation is that backward movement

may require more distributed recruitment of the locomotor network than forward movement. At the same time, the magnitude of this asymmetry may be influenced by modeling choices, including imposed binary command states, supplemented proprioceptive feedback channels, and the DMD-based criterion used to define loss of oscillation. We therefore regard the forward–backward difference as a model-generated prediction that should be tested against future circuit-level experiments, rather than as a resolved biological fact.

Hyperparameter tuning is critical for stable multi-objective optimization bounded by biophysical constraints

In our optimization process, the hyperparameters, which include coefficients of error terms, played a crucial role in controlling the trade-offs required to jointly capture the relative weight proportions in the anatomical connectome, the muscle activity patterns required for locomotion, and the phase relationship between SMD neuron activity and neck bending, all under the biophysical constraints of the model. Because the time-series-dependent terms (E_{BWM} and E_{SMD}) and the terms (E_{θ_w} , $E_{\theta_{w'}}$, and $E_{|w|}$) that act directly on synaptic weights affect optimization through qualitatively different mechanisms, coefficient tuning in this framework does not correspond to enforcing equal error contributions by each term. Nevertheless, whereas $E_{|w|}$ was included primarily to restrain the growth of synaptic-weight magnitudes rather than to decrease as much as the data-fitting terms, the other error terms were reduced during optimization, supporting that the empirically chosen coefficients enabled joint optimization without the process being effectively dominated by only one or two terms (Fig. S5). This study therefore asks whether the present framework admits a synaptic weight set within a functional space bounded by the leaky-integrator dynamics, the membrane potential time constant, and the topology of the anatomical connectome. Successful convergence indicates that the present framework can accommodate synaptic weight sets that jointly capture these targeted features under the biophysical constraints of the model.

Application of our strategy to other nervous systems

Our new strategy, featuring the combined form of error function and its minimization process, offers the advantage of easy applicability for other researchers. Researchers can custom-design objective-specific error terms according to their research needs and integrate it into the error function with an appropriate coefficient. These terms will assess if the activities of neurons, beyond SMD neurons, accurately reproduce experimental observations. By minimizing the error function with these additional terms, a more advanced model can be developed in the future.

Limitations

Despite our efforts, our research still presents several limitations. The reported Pearson correlations between anatomical and optimized synaptic weights should not be interpreted as evidence that all individual synaptic weights are reproduced with equal fidelity. Rather, these values summarize broad correspondence between the optimized and anatomical synaptic weight structures under connectome constraints. Because the model uses a fixed topology and simplified neuronal dynamics, the optimized parameters should be interpreted as model-dependent, task-specific effective synaptic weights rather than as uniquely identifiable biological synaptic strengths. In addition, the existence of multiple successful machine-learning trajectories indicates degeneracy in the solution space. It remains unclear whether this degeneracy arises from flexibility of the model, incomplete physiological constraints, or natural biological variability of the connectome across individuals^{5,7}. In particular, this degeneracy may reflect either biological robustness or redundancy in the *C. elegans* locomotor circuit or underdetermination of the present model due to limited physiological and behavioral constraints. We therefore do not interpret any single optimized trajectory as a unique biological solution. Instead, we regard features that recur across converged trajectories, such as broad correspondence with anatomical synaptic weights, appropriate locomotor kinematics, and repeated identification of particular neurons in ablation-based analyses, as more informative than any individual parameter set. Testing spiking neuron formulations remains an important future direction.

Second, experimental findings²⁹ indicated that *trp-1,2* mutation did not eliminate the oscillation of SMDD neuron activity, but merely altered its phase. However, *trp-1,2* mutation completely silenced the SMDD oscillation in the CANN model. A further limitation is that, although the model reproduced the qualitative locomotor bias associated with the simulated *trp-1,2* perturbation, the present analysis does not yet fully explain the internal dynamical mechanism by which removal of proprioceptive input to SMDD leads to circular locomotion. Third, because the forward- and backward-locomotion command artificially set membrane potentials of command-neurons to constant values of 0 or 1, the activity of command-neurons does not vary continuously, potentially diverging from the actual neuronal activity in *C. elegans*. Accordingly, command interneurons were excluded from the type I/II necessity analyses. The reported type I/II necessary neurons should therefore be interpreted as model-dependent elements among the non-command motor neurons, not as a complete biological necessity map of the locomotor circuit. Also our ablation analyses did not identify sufficient neurons. Sufficiency would require showing that the retained set alone can generate the oscillatory muscle pattern. Fourth, it is established that during forward (or backward) locomotion in *C. elegans*, the A-type (or B-type) motor neurons are not activated⁶⁰. Conversely, in the CANN model, the membrane potential of A-type (or B-type) motor neurons oscillated under forward-locomotion command (or backward-locomotion command), presenting a notable discrepancy. To the best of our knowledge, no studies have yet addressed this discrepancy. Fifth, we utilized the anatomical connectome data from prior work without questioning its reliability². However, this does not imply that we consider the data to be without any need for refinement. Sixth, we primarily constrained our model using behavioral data rather than large-scale population neural activity datasets^{11,12,15–19,61,62}. This choice was made because current whole-brain imaging techniques do not yet comprehensively capture the

motor neurons activity within the ventral nerve cord with neuron identification, which are essential for the locomotion tasks modeled in this study. Seventh, we supplemented proprioceptive connections that are absent from the anatomical connectome data and introduced them as effective couplings for modeling purposes. This limits direct biological interpretation of those specific synaptic weights. Eighth, the conversion from muscle contraction (r_i) to body control angle (θ_{ctrl}) requires a fixed scale factor in the present implementation. Because this factor is not independently derived from biophysical measurements, the absolute bending amplitude and the resulting locomotion speed in *ElegansBot* should be interpreted cautiously. Ninth, the optimization targets and out-of-objective validations in the present study were restricted to forward and backward locomotion. We therefore did not test whether an optimized weight set generalizes to other behaviors, such as turning or escape-like responses, without additional fitting. In addition, because the present behavioral readout is coupled to a two-dimensional biomechanical simulator, extending the model to more complex three-dimensional navigation would require an expanded neuromechanical formulation and separate validation.

Data availability

All data generated or analysed during this study are included in this published article, its Supplementary Information files, and the Zenodo repository, <https://zenodo.org/records/18626935>.

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Author contributions

TC and SK conceived the idea for this work. TC and SK set up the model, conducted the calculation, and analyzed the data and results. TC and SK wrote the manuscript. All authors reviewed and agreed on the contents of the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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